

Hybrid Polymer-Lipid Membranes as a Virtual Platform for Membrane Biology *Toward Engineering Membranes with Programmable Function*

Hybrid Polymer-Lipid Membranes (HM) are formed by co-self-assembly of an amphipathic block copolymer (e.g., polybutadiene-*b*-polyethylene oxide) and one or more biological lipids. Compared with pure lipid membranes, HMs show enhanced chemical and mechanical stability while still supporting the functional reconstitution of integral membrane proteins (IMPs). This unique combination makes HMs a versatile platform for advancing both technology and fundamental membrane biology.

Recently, we used coarse-grained Martini¹ simulations to identify two distinct mechanisms that facilitate protein incorporation in these membranes.^{2,3} Within a narrow compositional window, lipid-rich nanodomains form and proteins preferentially partition into these domains. Outside this region, we observe that hydrophobic mismatch around embedded proteins induces membrane defects that dynamically recruit lipids and displace the polymer, reducing the energetic cost of protein accommodation. This reorganization is coupled to the protein's conformational state and alters the local availability of individual lipid species. Using the synthetic membrane protein ROCKET, we show that this lipid enrichment can reshape preferential lipid-protein interactions while preserving established cardiolipin recognition. Local enrichment therefore promotes preferential binding but does not determine it on its own.

In future work, we will build on this structure-function relationship and use hybrid membranes as a platform to determine how local lipid concentration and molecular recognition jointly shape preferential lipid binding. By systematically tuning these factors, we aim to uncover the physical principles that generate lipid fingerprints around membrane proteins and ultimately use those principles to engineer membranes with programmable biological function.

References

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